Ruchita Kailas Vehale

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Education

University of Bristol Bristol, UK
Master of Science, Robotics Aug 2024 - Sep 2025
Relevant Coursework: Robotics Systems, Machine Vision, Artificial Intelligence, Robotic Fundamentals

Savitribai Phule Pune University (SPPU) Pune, India
Master of Technology, Electronics & Electrical **| GPA:** 9.43/10 (Gold Medallist) Jun 2020 - Sep 2022
Relevant Coursework: Embedded system and RTOS - ARM, Mechatronics and Transducer design, Machine Learning, Analytical Instrumentation, PLC, Communication Networks, Advanced Signal Processing
Design Project: Object Detection & Velocity Estimation using Digital Image Processing and Computer Vision.

AISSMS College of Engineering, SPPU Pune, India
Bachelor of Engineering, Electronics Engineering **| GPA:** 8.31/10 May 2016 - Nov 2020
Relevant Coursework: Microcontroller and Applications, VLSI, Embedded Processes, Analog and Digital Electronics, IoT, Wireless sensor network, Digital Electronics and FPGA, Signal Processing, Advanced Power Electronics
Design Project: Text to Speech synthesis in Celebrity's voice.

Experience

DeGould Ltd Bristol, UK
Associate Software Engineer (R&D Team) October 2025 – Present

- Developed end-to-end localisation solutions for automotive defect detection systems, ensuring accurate identification and positioning of defects across multiple vehicle inspection pipelines
- Improved system consistency and maintainability by refining codebase structure and version control workflows across the team
- Designed and implemented computer vision and machine learning algorithms for calibration, pose estimation, and defect localisation, enhancing automation and reducing manual intervention
- Contributed to scalable deployment of inspection services by optimising performance and reliability in production environments and eventually making faster deployments

HVN Labs, Future Space, BRL Bristol, UK
Jr. Drone System Engineer June 2025 – August 2025

- Successfully implemented a reverse landing system for drones using GPS and Raspberry Pi camera, enabling precise autonomous landings.
- Currently developing a stereo camera-based system to improve depth estimation and enhance visual navigation capabilities.
- Also working on LiDAR and camera sensor fusion for reliable drone detection and accurate landing, with Wi-Fi communication and MAVLink protocol integration for seamless control and data transmission.

HVN Labs, Future Space, BRL Bristol, UK
Technology Intern April 2025 – May 2025

- Currently part of the R&D and build team for developing a coaxial drone system intended for synchronized lighting displays. Involved in drone assembly, control systems, and performance testing.

- Also working on a precision autonomous landing system using sensor fusion of GPS and camera data to enhance landing accuracy and reliability in dynamic environments.

CMET Pune, India

Research Fellow April 2023 - Aug 2024

- Optimized antenna designs with CST Microwave Studio, achieving 20% size reduction and 70% improved accuracy for GPS, Wi-Fi, Bluetooth and 5G frequencies.
- Operated 3D printers (FDM, Inkjet) and characterization tools (XRD, VNA) for antenna prototyping and testing.
- Fabricated a biodegradable-ink strain sensor via 3D inkjet printing.
- Worked on 3D printed coin cell Batteries.

HEMRL, DRDO

Technology Intern Nov 2021 - Aug 2022

Project: Object detection & velocity estimation using Digital image processing & Computer vision**.**

- Developed and deployed a MATLAB based video-based tracking, incorporating moving object recognition using Gaussian Mixture Model and point tracking algorithm, achieving higher accuracy in variable lighting conditions.
- Executed feature extraction and depth estimation employing projective geometry across multiple planes for precise velocity measurement across 20 video data set, showcasing problem solving skills.

Projects

1. Magnet-Detecting Robot Using Polulu 3Pi+

- Programmed a robot with boundary detection, PID control, and calculative path planning while implementing shortest path finding algorithm for accurate magnet location identification.

2. Leader-Follower Robot Using QTR-RC Sensors

- Created a robotic system with dynamic speed control and sensor fusion, reducing tracking errors by 97.93% for autonomous and swarm robotics applications, and a wrote a paper on it.

3. CIFAR-10 Object Detection

- Implemented deep learning with Keras to classify CIFAR-10 images, presenting results in an intuitive grid for analysis.

4. Hybrid Apple Detection System

- Combined segmentation techniques with YOLO V8 with 1500 epochs which achieving accuracy of 98.87% in apple detection while improving processing efficiency and compared it with traditional machine vision algorithm.

5. Serial and Parallel Robot Kinematics

- The project explored Robot Kinematics, featuring the LynxMotion arm with forward and inverse kinematics, trajectory plotting, and obstacle avoidance, and parallel robot for surgical operations simulated in MATLAB.

6. Human-Robot Interaction (HRI) Project – NAO Robot Language Tutor

- Designed and implemented an AI-driven NAO robot language tutor with real-time speech interaction, expressive gestures, and interactive learning features using socket programming and choregraphe.

7. Artificial Intelligence

- Evaluated Supervised, Unsupervised and Reinforcement Learning algorithms on Dungeons and Dragons maze dataset. Developed a complete maze game by integrating all three approaches, utilizing the same dataset and visualized the results in Pygame environment.

8. Dynamic Navigation and collaborative mapping

- Developed a dynamic object-aware visual SLAM system integrating YOLOv8, DeepSORT, and MiDaS for real-time detection, tracking, and depth estimation
- Implemented RGB and depth-based dynamic filtering with ORB-SLAM and Open3D, improving mapping accuracy

in dynamic environments

- Achieved robust localisation and mapping with reduced dynamic object interference and real-time performance across monocular and RGB-D systems

Publications

1. **Binder Jet 3D Printed Ceramic Substrate for 5G Patch Antenna**
   Ruchita K. Vehale, *et al.*, *Advanced Materials Technologies (Wiley)*

2. **Comparative Analysis of Conventional and Deep Learning Algorithms for Apple Detection**
   Ruchita K. Vehale, *et al.*, *SSRN Electronic Journal*

3. **Preparation of Phase-Pure $MgTiO_3$ Microwave Dielectric Ceramics for GPS Antenna Application**
   Ruchita K. Vehale, *et al.*, *Journal of Materials Chemistry C*, January 2024

4. **Correction: Preparation of Phase-Pure $MgTiO_3$ Microwave Dielectric Ceramics for GPS Antenna Application**
   Ruchita K. Vehale, *et al.*, *Journal of Materials Chemistry C*

5. **Preparation of ABS-ZnO Dielectric Nanocomposite, its Filament and Fabrication of 3D Printed Microstrip Patch Antenna GPS Module for Communication Application**
   Ruchita K. Vehale, *et al.*, *Preprint*, January 2025

6. **Hierarchical Quantum-Accelerated Federated Learning for Scalable, Auditable Cross-Enterprise AI Governance**
   Ruchita K. Vehale, *et al.*, *IJSRET*

Skills and Certifications

- **Programming & Tools:** Python, C/C++, Embedded C, MATLAB, Verilog, Git, AWS
- **Machine Learning & Computer Vision:** OpenCV, NumPy, Scikit-learn, Keras, PyTorch (model usage), YOLOv8, DeepSORT, ORB-SLAM
- **Design & Simulation:** Blender, SolidWorks, AutoCAD, GIMP, CST Studio
- **Certifications:** Python for Deep Learning Applications; Mechatronics in PLC and SCADA
- Ongoing Machine Learning and Artificial Intelligence course – Harvard University
- Experienced with Bot Labs Software and Microsoft Office Suite
- Led robotics workshops and received awards in robotics competitions

Leadership & Activities

National Cadet Corps (3 Mah Air Squadron) Pune, India
Cadet Seargent May 2017 - Feb 2020

- Exhibited leadership and management skills for engagement in military activities, earning B & C Certificates.
- Awarded the Best Cadet Trophy for leading a team of 500 student at camp, delivering training and Guidance.
- Earned a Gold Medal in 0.22 Rifle Shooting at the NCC Camp, demonstrating precision and marksmanship skills.

Activities: Competed in inter-college and national robotics competitions as a member of the Robotics and Drone Club, while also completing a two-year soft skills training program and attending a leadership development workshop.
Language: English, Hindi, Marathi, Japanese (N4 &N5 level)
Interests: Tabla - 5 levels (Classical instrument), Tree plantation, Rifle shooting, Painting